

# Computer Science & IT

## Discrete Mathematics



### Graph Theory

Lecture No. 07



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# Recap of Previous Lecture



Topic

Simple connected planar graph



Topic

Euler's equation for connected planar graph



Topic

Polyhedral graph



Topic

Euler's equation for disconnected planar graph



Topic

Vertex coloring



# Topics to be Covered



✓  
Topic

Matching

✓  
Topic

Line covering

✓  
Topic

Line independent set

✓  
Topic

Vertex covering

✓  
Topic

Vertex independent set

Matching





## Topic : Matching

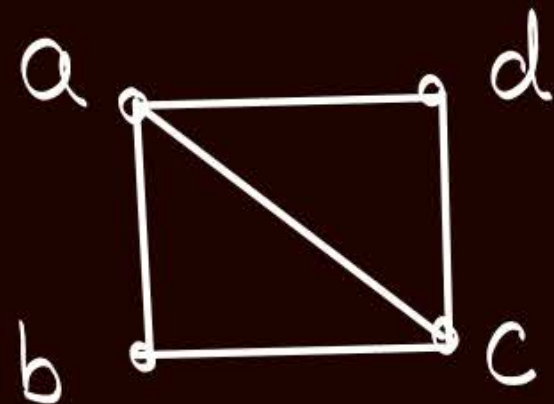


A subgraph  $\overset{\checkmark}{M}$  of graph  $\overset{\checkmark}{G}$  is called a matching of graph  $G$ , if all the vertices of graph  $G$  are incident with at most one edge in subgraph  $M$ ,

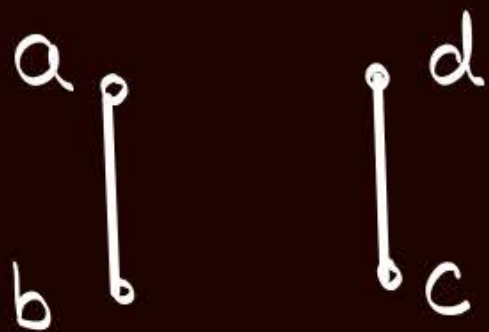
↳ All vertices of graph  $G$  must be present in sub-graph  $M$

In a Matching of graph  $G$ ,  $\deg(v) \leq 1 \quad \forall v \in G$

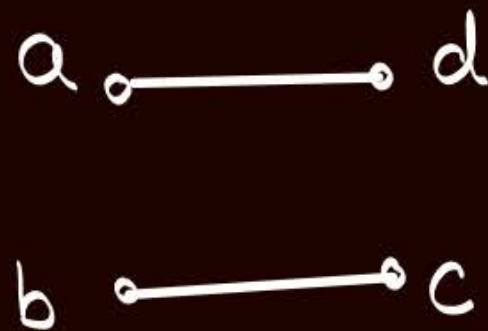
eg:



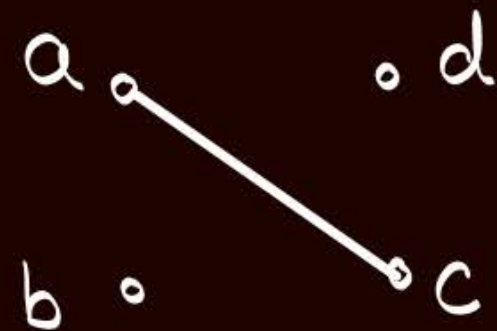
$G$



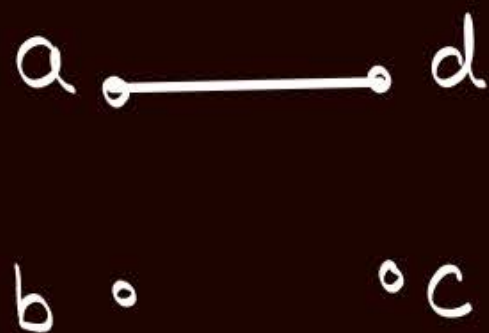
" $M_1$ "



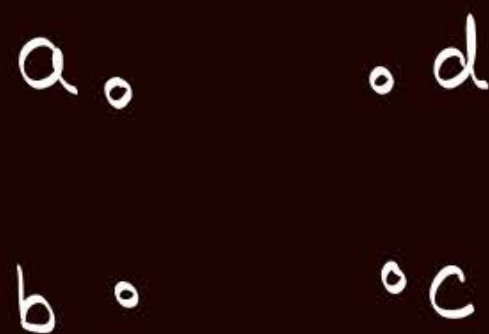
" $M_2$ "



" $M_3$ "



" $M_4$ "



" $M_5$ "

In a matching of graph  $G$ , adjacent edges are not allowed.

$M_1, M_2, M_3, M_4, M_5$ , all are matching of graph  $G$ .  
Other matchings are also possible for graph  $G$ .

Note:

- ① In a Matching of graph  $G$ .  
if  $\deg(v) = 1$ , then vertex ' $v$ ' is called a matching vertex
- ② In a Matching of graph  $G$ .  
if  $\deg(v) = 0$ , then vertex ' $v$ ' is called a non-matching vertex



Maximal



Nothing can be  
added, without  
destroying the  
Property

Minimal



Nothing can be  
removed, without  
destroying the  
Property





## Topic : Maximal Matching

- A matching of graph  $G$  is called a maximal matching, if no other edge of graph  $G$  can be added to that matching without destroying its property of being a matching.

In above example,

$M_1$ ,  $M_2$  and  $M_3$  are maximal matching of graph  $G$ .





## Topic : Maximum Matching

## Largest Maximal Matching

- matchings of graph  $G$  with maximum number of edges of graph  $G$ ,  
are called maximum matching.  
{ There may be more than one maximum matching for the given graph }

In the above example,

$M_1$  &  $M_2$  are maximum matching

{  $M_3$  is maximal matching,  
but not a maximum matching }





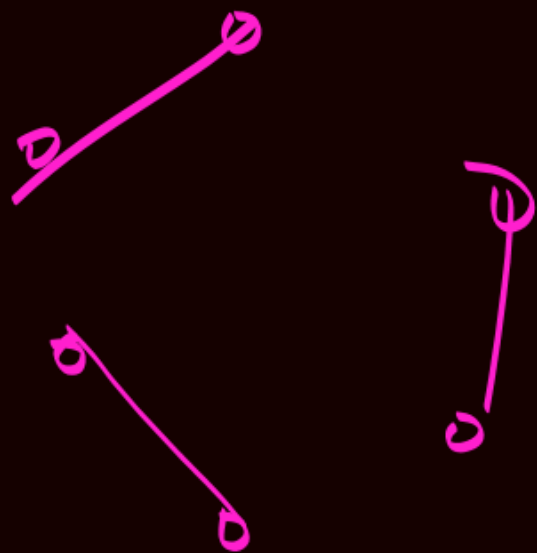


★ Matching Number of graph  $G$  = Number of edges in any one of the maximum matching of graph  $G$ .

In the above example,

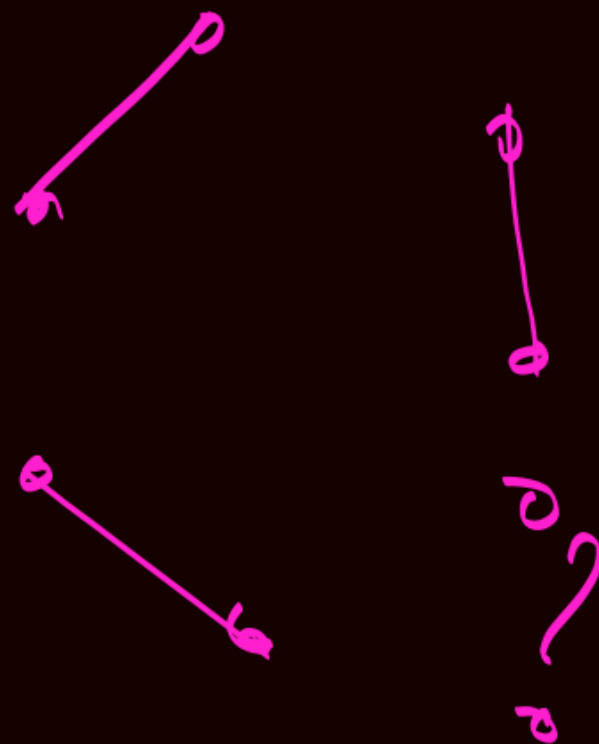
Matching no. of  $G = 2$   
 $\uparrow$   
 w.r.t.  $M_1$  or  $M_2$

$n$ -Even  $n$



$$\frac{n}{2} \text{ edges} \Rightarrow \left\lfloor \frac{n}{2} \right\rfloor$$

$n$ -odd



$$\left( \frac{n-1}{2} \right) \text{ edges}$$



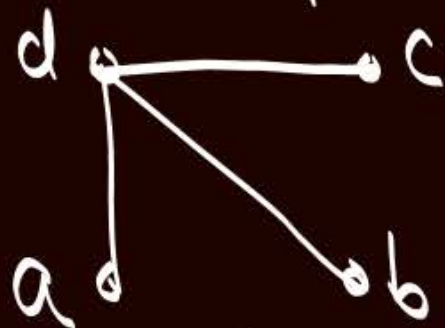
Note:-

Let  $G$  be a graph with  $n$ -vertices,  
then

matching number of graph  $G$  is  $\leq \lfloor \frac{n}{2} \rfloor$

↑  
it is the upper bound on  
matching No.

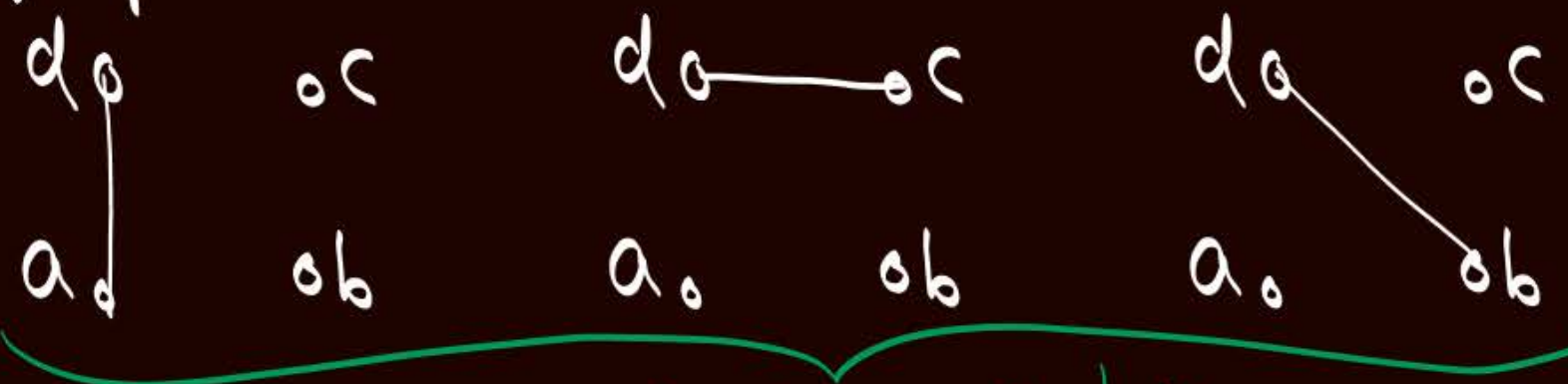
eg: Consider the following graph



$G$

in graph  $G$ ,  
 $n=4$

graph



All are maximum  
of graph  $G$

∴ Matching  
No. of  $G = 1 < \lfloor \frac{4}{2} \rfloor$

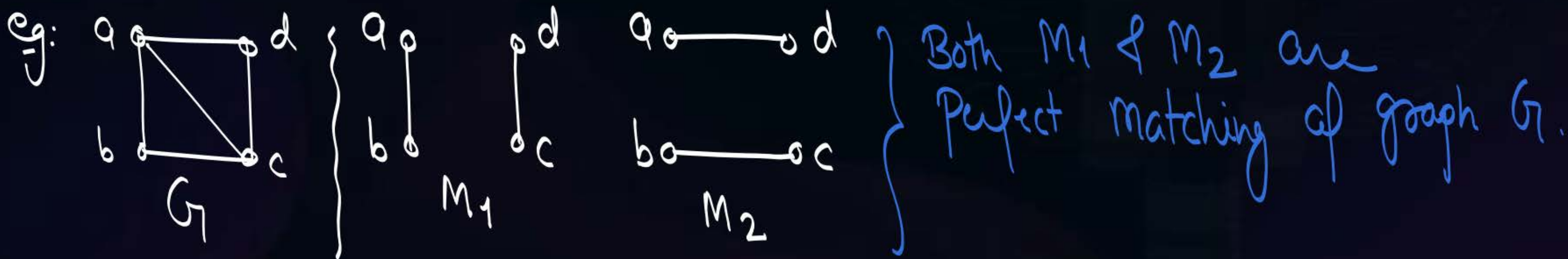


## Topic : Perfect Matching



A matching of graph  $G$  is called a perfect matching if and only if degree of each vertex in the matching of graph  $G$  is '1'.

in a perfect matching,  $\deg(v) = 1, \forall v \in G$

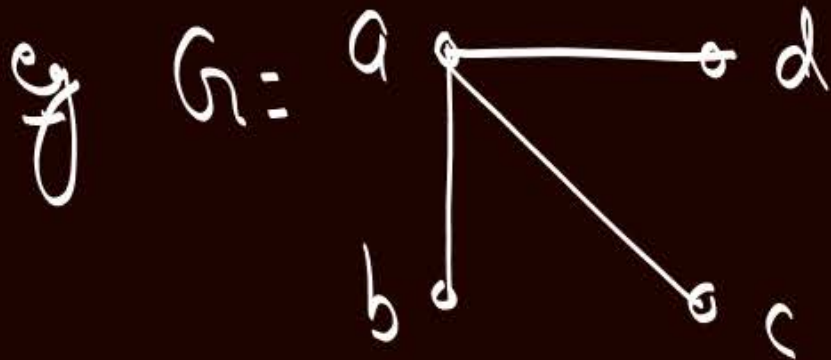




Note :-

① Every perfect matching of graph  $G$ , is also a maximum matching of graph  $G$ .

② But, Every maximum matching of graph  $G$  need not be a perfect matching



} Maximum matching exist for graph  $G$ , but no perfect matching



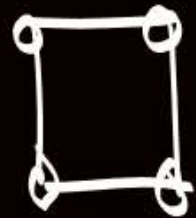
Note:-

① Perfect matching need not exist for Every graph

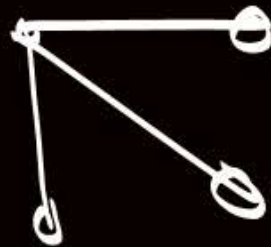
② In a graph  $G$ ,  
If no. of vertices are odd, then perfect matching  
Can never exist for that graph

③ In a graph  $G$ ,  
If no. of vertices are even, then Perfect matching  
may or may not exist for that graph

eg.



eg.



Note:-

If perfect matching exists for graph  $G$ ,  
then number of vertices in graph  $G$  are even.  
(Converse of the statement need not be true)

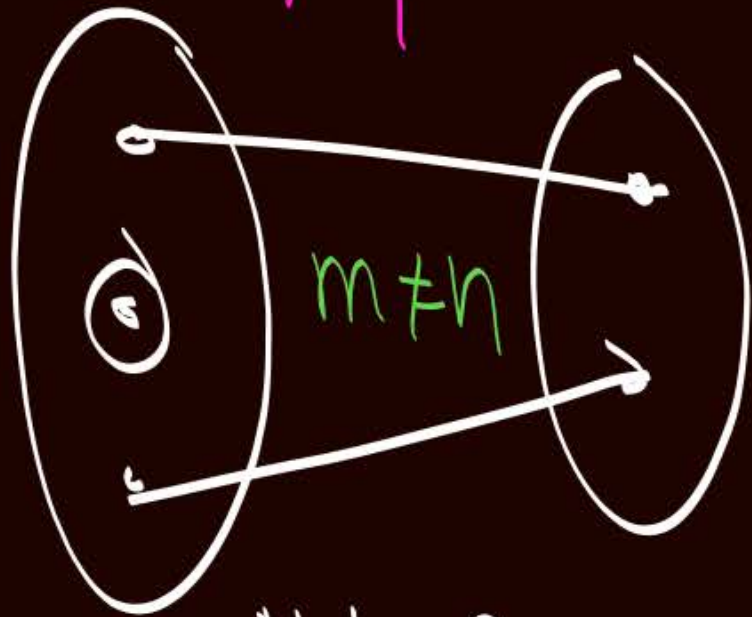
Note:-

Let  $G$  be a graph with  $n$ -vertices,  
If perfect matching exists for graph  $G$ , then  $n = \text{even}$ .  
And Matching Number of graph  $G = \frac{n}{2}$

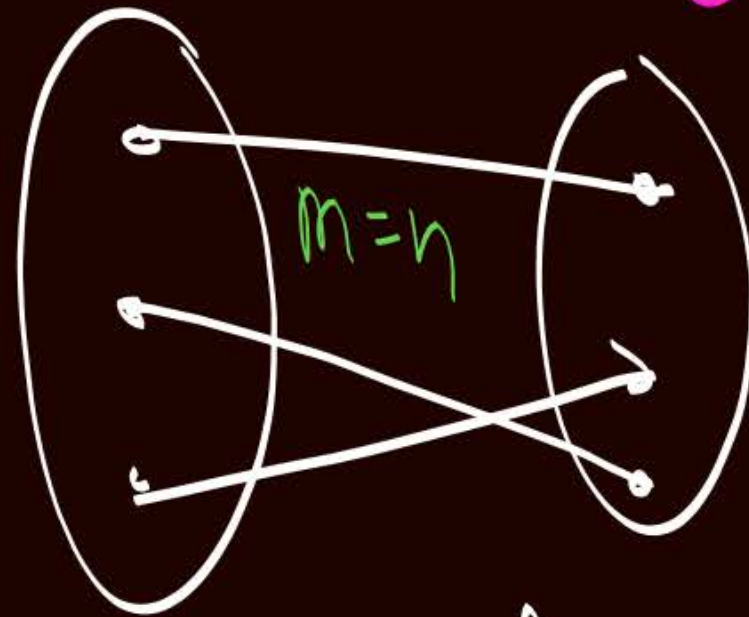


④ For a Complete graph  $K_n$  perfect matching exists if and only if  $n = \text{Even}$  in  $K_n$

④ In a Complete bipartite graph  $K_{m,n}$  perfect matching exist if and only if  $m = n$

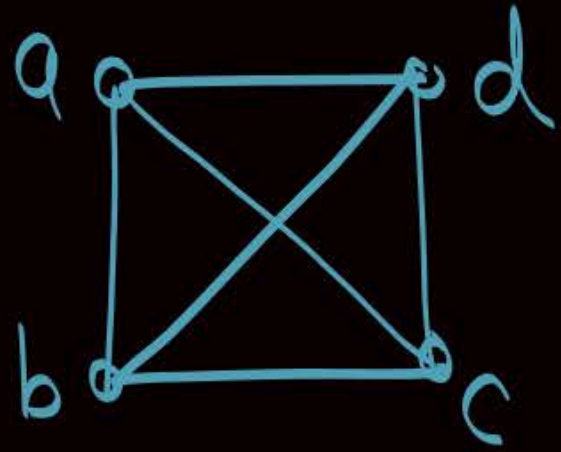


Not a  
perfect matching

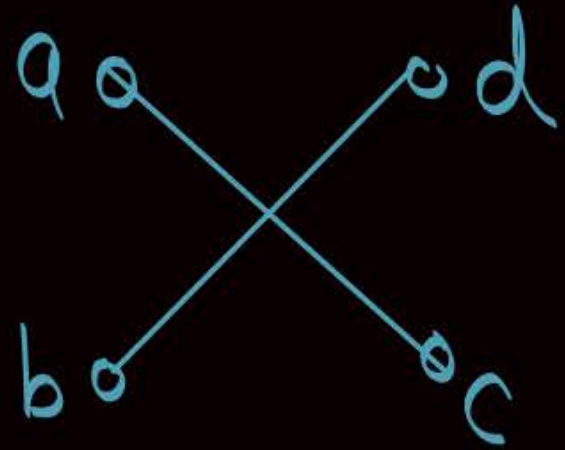
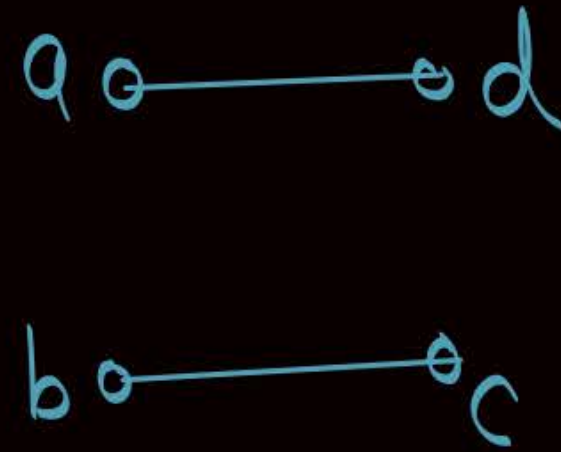
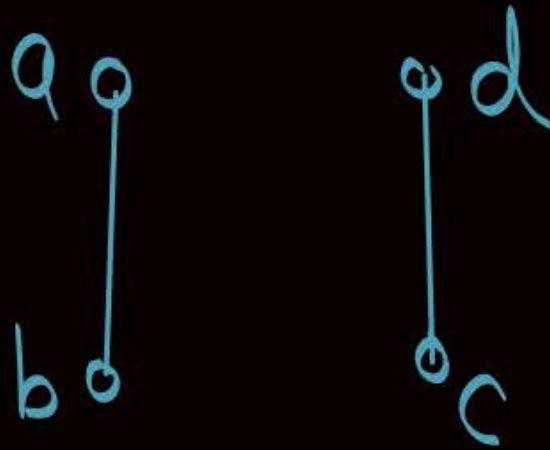


it is perfect matching

\* Let  $K_n$  is a Complete graph, where ' $n$ ' is Even.  
Find the no. of Perfect matching in Complete graph  $K_n$



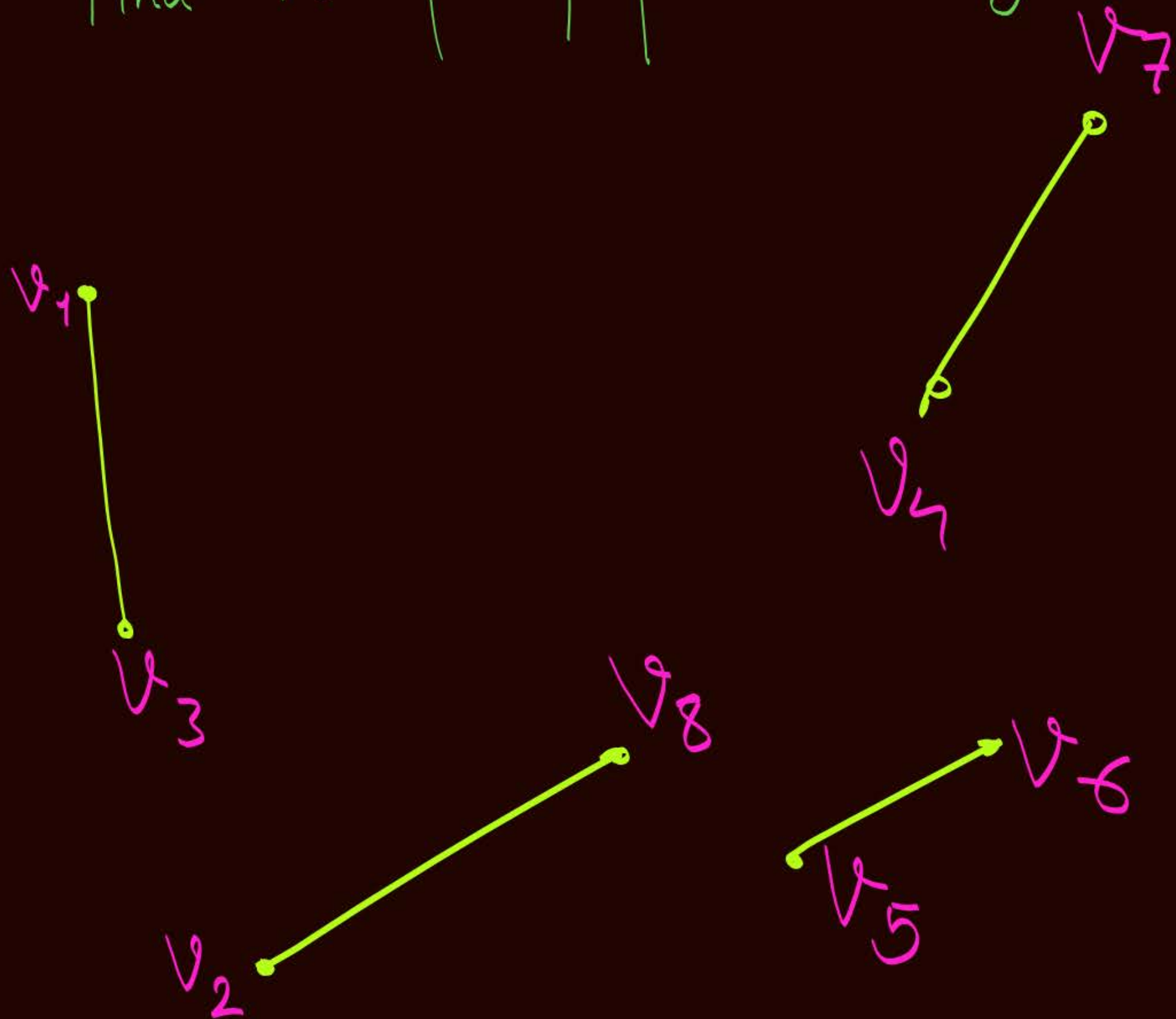
$K_4$



These are the only three perfect matchings possible for  $K_4$



Find no of perfect matchings in  $K_8$



$$\begin{matrix} (8-1) & (8-3) & (8-5) & (8-7) \\ 7 & \times 5 & \times 3 & \times 1 = 105 \end{matrix}$$

# Number of Perfect matching in $K_n$ $\{n=\text{Even}\}$

Number of  
Perfect matching  
in  $K_n$

$$= (n-1)(n-3)(n-5) \dots 5 \cdot 3 \cdot 1$$

$$= (n-1)(n-3)(n-5) \dots 5 \cdot 3 \cdot 1 \cdot \frac{n(n-2)(n-4) \dots 6 \cdot 4 \cdot 2}{n(n-2)(n-4) \dots 6 \cdot 4 \cdot 2}$$

$$= \frac{n(n-1)(n-2)(n-3) \dots 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{n(n-2)(n-4) \dots 6 \cdot 4 \cdot 2}$$

$$= \frac{n!}{(2 \cdot \frac{n}{2}) \cdot (2 \cdot (\frac{n}{2}-1)) \cdot (2 \cdot (\frac{n}{2}-2)) \dots (2 \cdot 3) \cdot (2 \cdot 2) \cdot (2 \cdot 1)}$$

$$= \frac{n!}{(2)^{n/2} \cdot \left\{ (\frac{n}{2}) \cdot (\frac{n}{2}-1) \cdot (\frac{n}{2}-2) \dots 3 \cdot 2 \cdot 1 \right\}} =$$

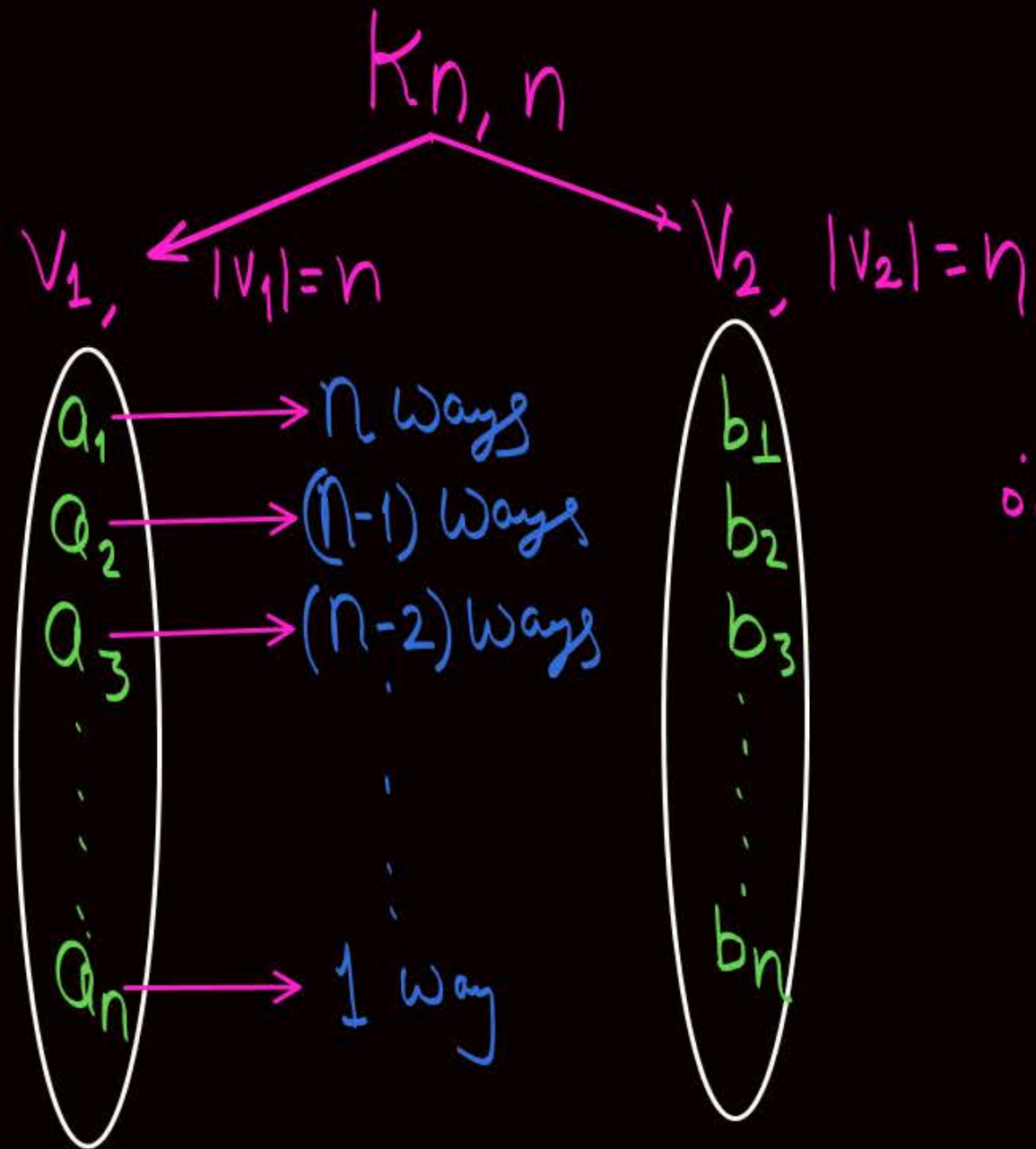
$$\frac{n!}{(2)^{n/2} \cdot (\frac{n}{2})!}$$



∴ Number of perfect matching in  $K_{2n}$   it will always be even  
just replace 'n' by '2n'

$$\therefore \text{Number of Perfect matching in } K_{2n} = \frac{(2n)!}{(2)^{\frac{2n}{2}} \cdot \left(\frac{2n}{2}\right)!} = \boxed{\frac{(2n)!}{2^n \cdot n!}}$$

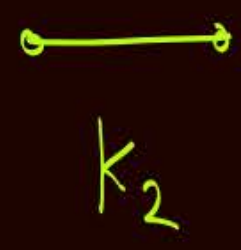
Q:- Find the number of perfect matching  
in Complete bi-partite graph  $K_{n,n}$



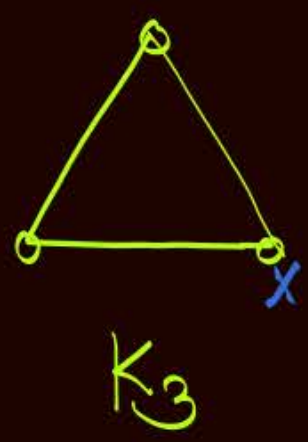
$\therefore$  # Perfect matchings  
in Complete bipartite  
graph  $K_{n,n}$  =  $n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$   
=  $n!$



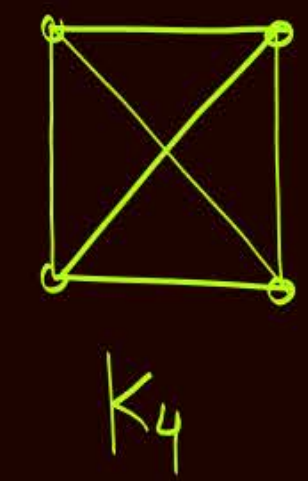
H.W. Q:- Find the matching number of Complete graph  $K_n$



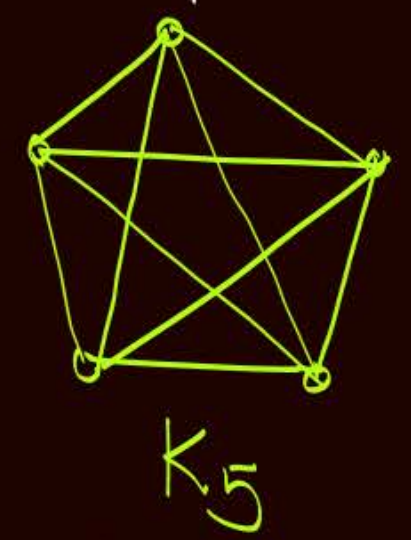
Matching No.  
of  $K_2 =$



Matching No.  
of  $K_3 =$

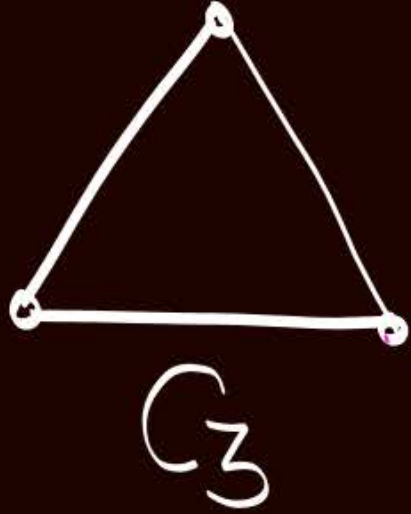


Matching No.  
of  $K_4 =$

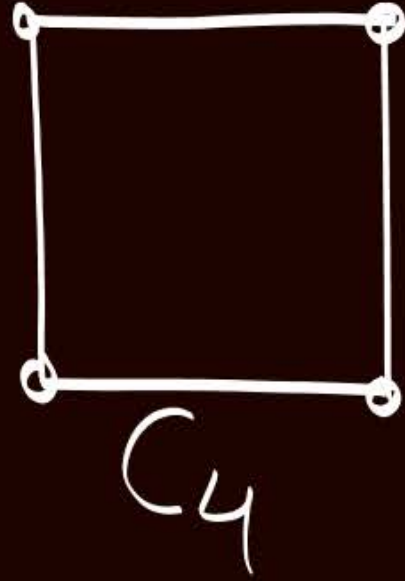


Matching No.  
of  $K_5 =$

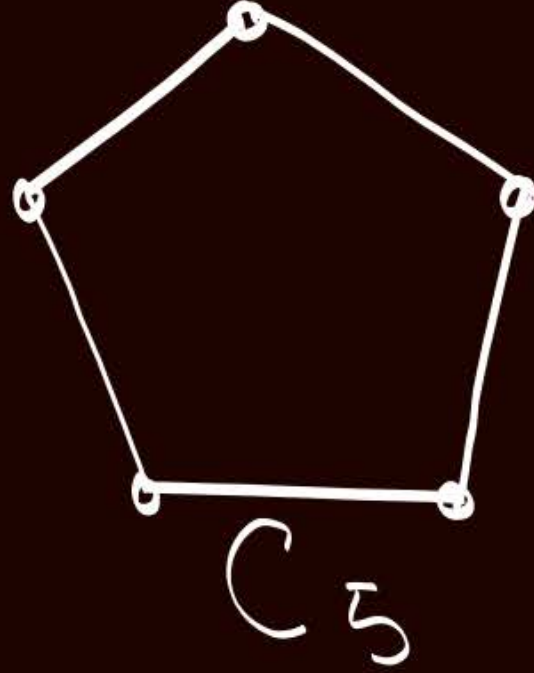
H.W.  
Q: Find matching number of Cycle graph  $C_n$



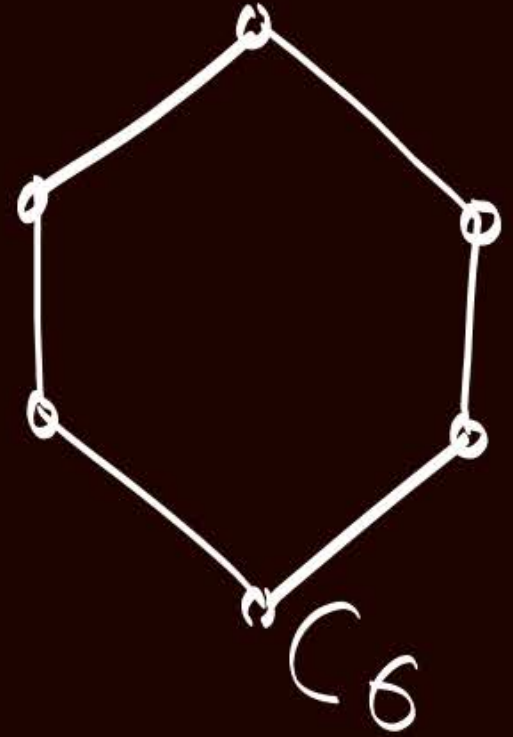
Matching No. =



Matching No. =



Matching No. =

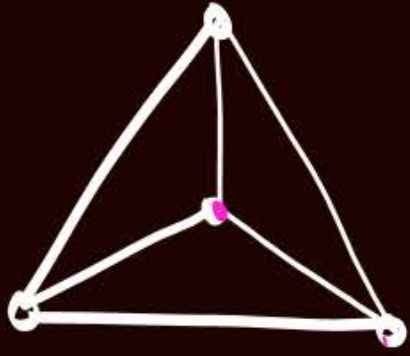


Matching No. =



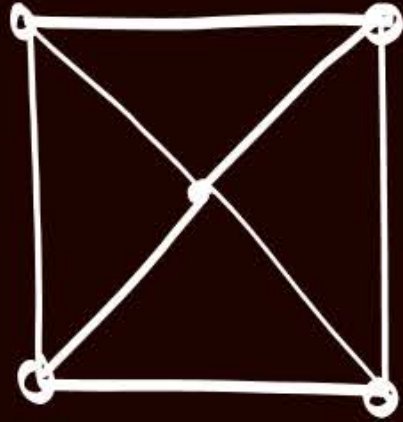
H.W.

Q: Find matching number of wheel graph  $W_n$



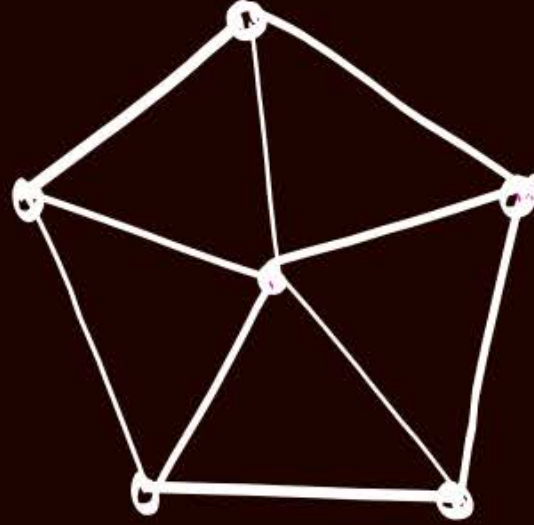
$$W_4 = C_3 + 1$$

Matching No. =



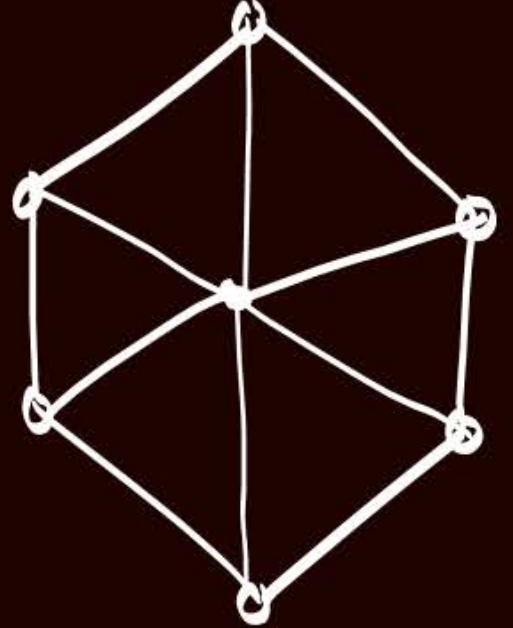
$$W_5 = C_4 + 1$$

Matching No. =



$$W_6 = C_5 + 1$$

Matching No. =



$$W_7 = C_6 + 1$$

Matching No. =

H.W

Q:- Find the matching number of Complete bipartite graph  $K_{m,n}$



Q:- Find the matching number of star graph with ' $n$ ' vertices.

# Covering

Line Covering / Edge Covering ✓

↓  
We will try to cover all the vertices of the graph using subset of edges

Vertex Covering ✓

↓  
We will try to cover all the edges of the graph using subset of vertices



## Topic : Line/Edge Covering

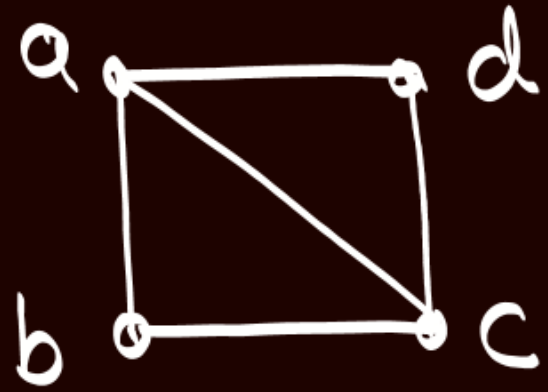
Let  $G=(V,E)$  be a graph.

A subset C of set of edges E is called line covering of graph G if every vertex of the graph is incident with at least one edge of subset C.

ie, in a sub-graph corresponding to the line covering  
 $\deg(v) \geq 1, \forall v \in G$



eg:-

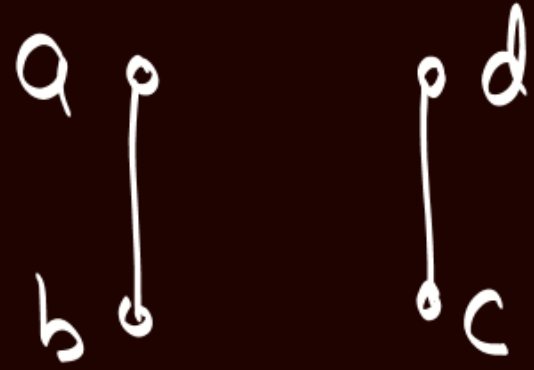


"G"

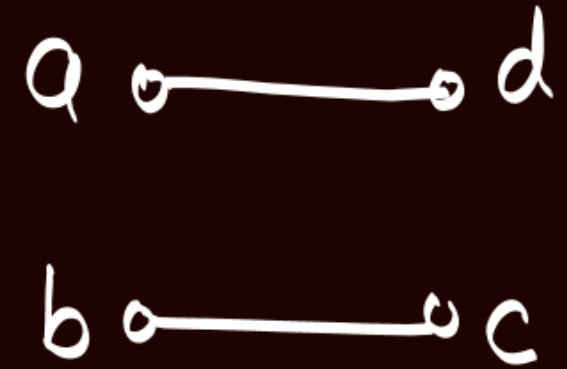
$$V = \{a, b, c, d\}$$

$$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}\}$$

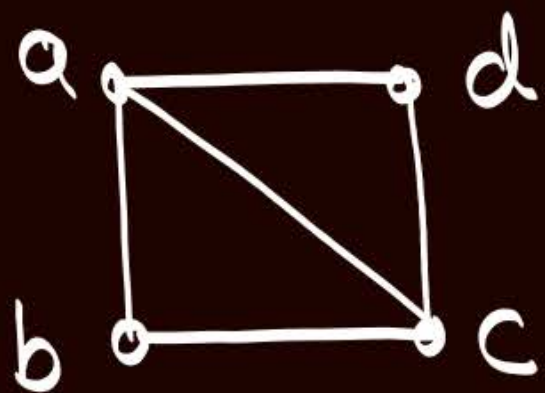
$$C_1 = \{\{a, b\}, \{c, d\}\}$$



$$C_2 = \{\{a, d\}, \{b, c\}\}$$



eg:-

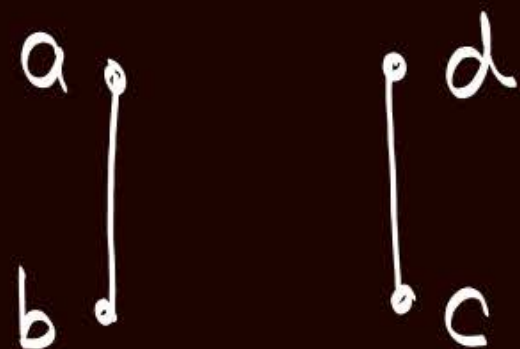


"G"

$$V = \{a, b, c, d\}$$

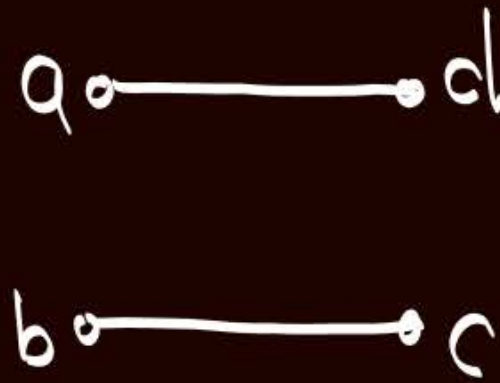
$$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}\}$$

$$C_1 = \{\{a, b\}, \{c, d\}\}$$

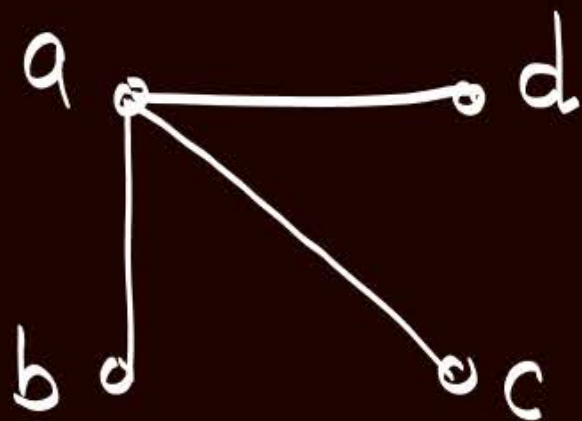


"C<sub>1</sub>"

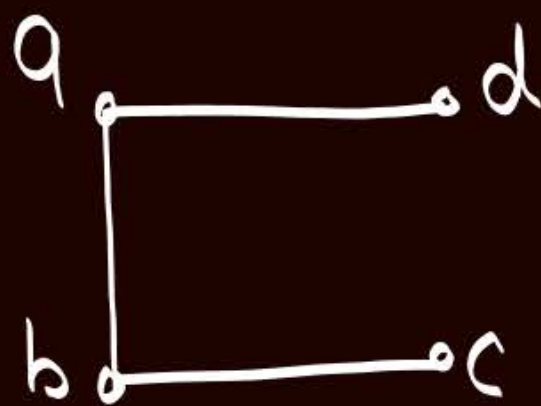
$$C_2 = \{\{a, d\}, \{b, c\}\}$$



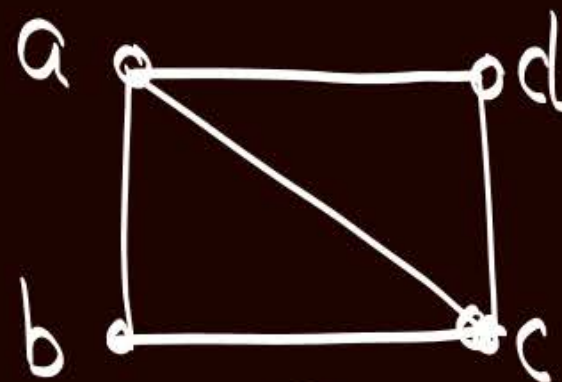
"C<sub>2</sub>"



$$C_3 = \{\{a, b\}, \{a, c\}, \{a, d\}\}$$



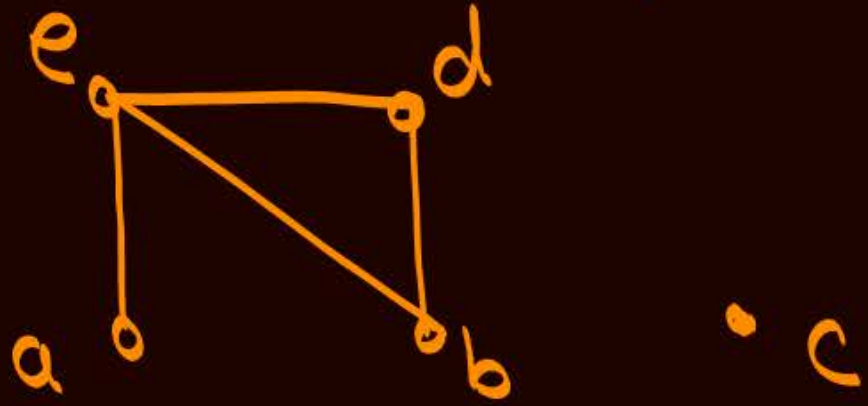
$$C_4 = \{\{a, b\}, \{b, c\}, \{a, d\}\}$$



$$C_5 = E$$

C<sub>1</sub>, C<sub>2</sub>, C<sub>3</sub>, C<sub>4</sub>, C<sub>5</sub>  
etc. are line  
covering of  
graph G

→ Consider the following graph  $G$



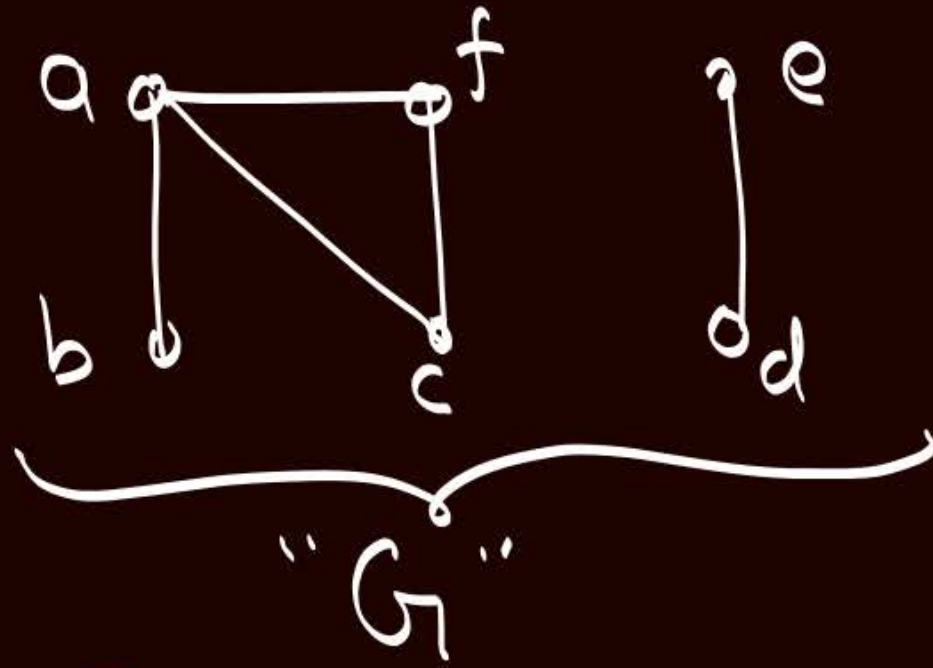
do we have any line covering for graph  $G$ .

Ans: No line covering exist for the given graph

Line covering does not exist for a graph  $G$   
if and only if any isolated vertex exist  
in graph  $G$



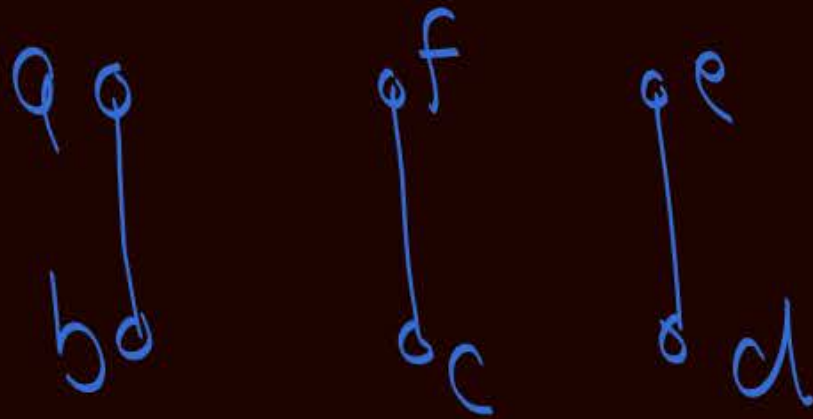
eg:



← disconnected graph

Does line covering exist for graph  $G$ .

Ans = Yes.



Line covering can exist for a disconnected graph as well, provided no isolated vertex exist.



## Topic : Minimal line covering

A line covering of graph  $G$  is said to be minimal if no edge can be deleted from that line covering without destroying its property to cover all the vertices of the graph.

In the above example,

$C_1, C_2$  &  $C_3$  are minimal line covering





## Topic : Minimum line covering

## Smallest minimal line covering



- Line coverings of graph  $G$  with minimum number of edges are called minimum line coverings of graph  $G$ .

In the above example,

$C_1$  &  $C_2$  are minimum line covering

$C_3$  is minimal, but not minimum



Note:-

- ① Every minimum line covering of graph  $G$  is also a minimal line covering, but every minimal line covering need not be a minimum line covering {eg.  $C_3$ }
- ② Minimal line covering of graph  $G$  can never contain any cycle
- ③ In a minimal line covering of graph  $G$  there can not be any path of length '3' or more





## Topic : Line covering number ( $\alpha_1$ )

Line covering no. of graph  $G$  = No. of edges in any one of the minimum line covering

$\alpha_1$  = No. of edges in any one of the minimum line covering

In the above example,  
line covering No. of  $G = 2$   
with  $C_1$  or  $C_2$

For any graph  $G$  with  $n$ -vertices  
line covering No. of  $G \geq \lceil \frac{n}{2} \rceil$





## Topic : Line independent set

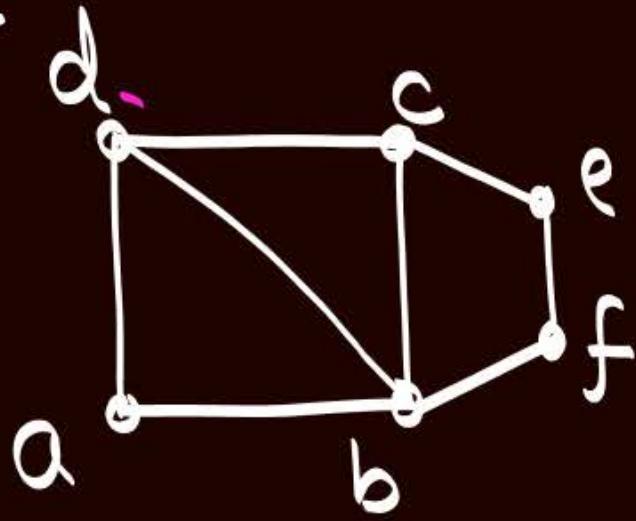
Let  $G = (V, E)$  be a graph.

A subset 'L' of set of edges 'E' is called a line independent set if no two edges of subset L are adjacent to each other.

Because in a line independent set no two edges can be adjacent  
∴ In a subgraph corresponding to line independent set,  $\deg(u) \leq 1, \forall u \in G$   
We know  $\deg(u) \leq 1, \forall u \in G$  is also the condition for matching  
Hence Every line independent set of graph G is also a matching of graph G.



eg:



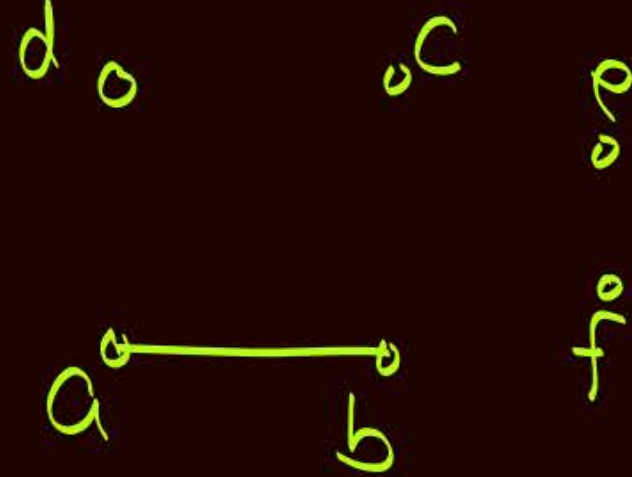
$G$

$$L_1 = \{ \{a, b\} \}$$

A subset of edges with a single edge is always a line independent set.

$$L_2 = \{ \{a, b\}, \{c, d\} \}$$

Matching w.r.t  $L_1$  =



$$L_3 = \{ \{a, d\}, \{e, f\} \}$$

$$L_4 = \{ \{b, d\}, \{c, e\} \}$$

$$L_5 = \{ \{a, d\}, \{b, c\}, \{e, f\} \}$$

$$L_6 = \{ \{a, b\}, \{c, d\}, \{e, f\} \}$$

All subsets of edges from  $L_1$  to  $L_6$  are line independent sets of graph  $G$ .



## Topic : Maximal Line independent set

- \* A Line independent set is called a maximal line independent set if no new edge from the graph can be added to the set without destroying its property of being independent set.

In the above example,

$L_4, L_5$  &  $L_6$  are maximal line independent set





## Topic : Maximum Line independent set

Largest maximal  
line independent set

- \* Line independent sets with maximum number of edges are called maximum line independent sets.

In the above example,

$L_5$  &  $L_6$  are maximum line independent set

→ Every maximum line independent set is also a maximal line independent set, but Every maximal line independent set need not be a maximum line independent set. { eg.  $L_4$  }





## Topic : Line independence number ( $\beta_1$ )

Line independence Number of graph  $G$  = Number of edges in any one of the maximum line independent set of  $G$ .

$\beta_1$  = No. of edges in any one of the maximum line independent set = Matching No. of graph  $G$

For a graph  $G$  with ' $n$ ' vertices  
Line independence no. of graph  $G \leq \lfloor \frac{n}{2} \rfloor$

In the above example,

Line independence No. of graph  $G = 3$

Wrt.  $L_5$  or  $L_6$



✓ \*

For any graph  $G = (V, E)$

$$\alpha_1 + \beta_1 = |V(G)|$$

i.e. (Line covering number of graph  $G$  + Line independence number of graph  $G$ ) = Total number of vertices in graph  $G$

Find  $\alpha_1$  ,  $\beta_1$

for the following graphs

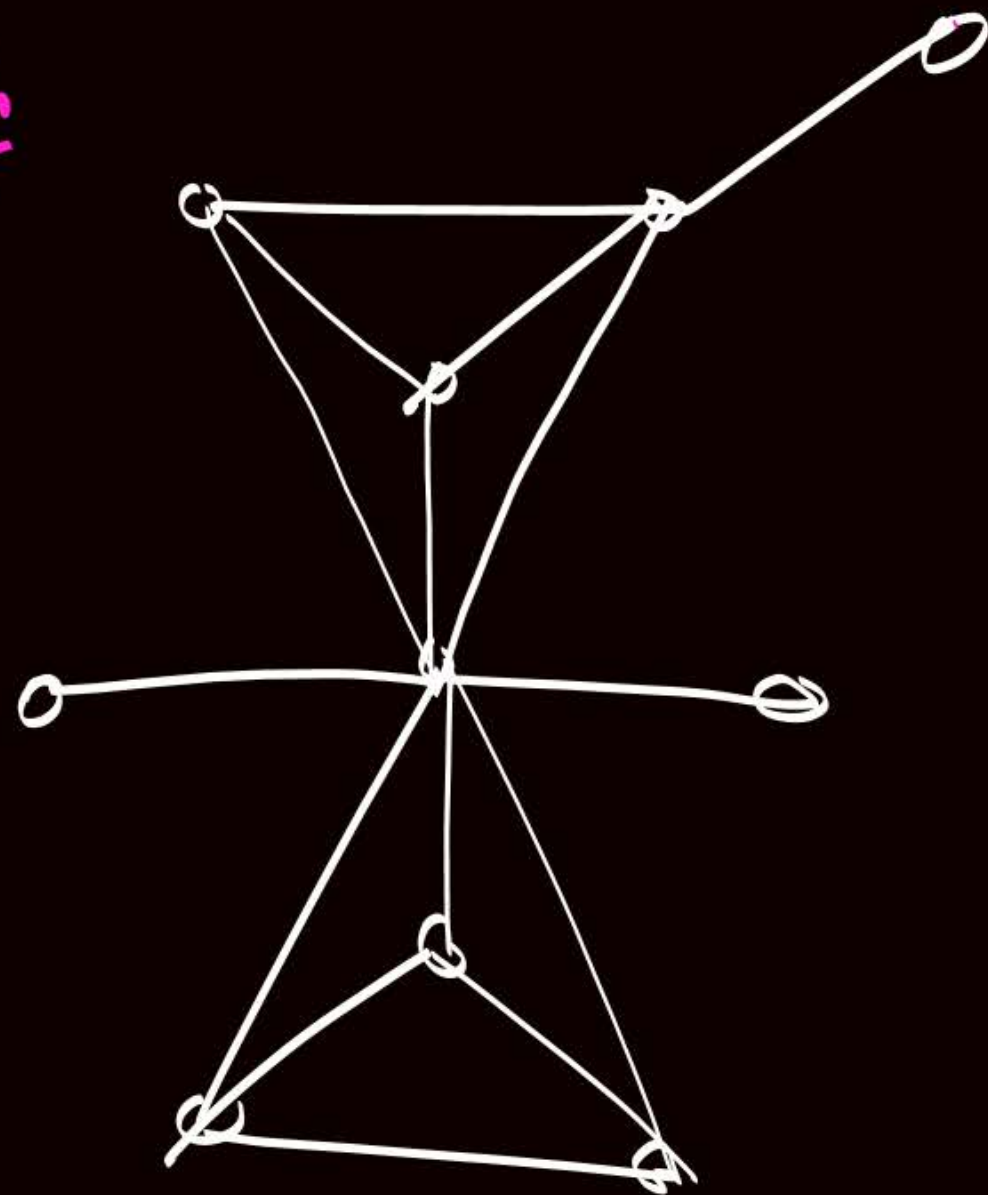
- (i) Complete graph  $K_n$
- (ii) Cycle graph  $C_n$
- (iii) Wheel graph  $W_n$
- (iv) Complete bipartite graph  $K_{m,n}$
- (v) Star graph with 'n' vertices

$\alpha_1$  = line covering no.

$\beta_1$  = line independence no.



Q.11



Find  $\alpha_1$  &  $\beta_1$  for given graph

$$\alpha_1 = ?$$

$$\beta_1 = ?$$



## 2 mins Summary



Topic

Matching

Topic

Line covering

Topic

Line independent set

Topic

Vertex covering

Topic

Vertex independent set



**THANK - YOU**